

Instructions:

- **Due May 12, 11:59PM uploaded to Blackboard as a single pdf.**
- Each problem is worth $\frac{100}{\text{number of problems}}$ points.
- Start each problem on a new page and put a box around your final answer to help the grader.
- Label your file as `ENME403_S26_HW11_yourname.pdf`.
- If you use Simulink to generate results for the homework submission, include screenshots of the block diagrams and all relevant plots.
- **Use of AI Tools:** The use of AI software to prepare your submission is strictly prohibited. If we determine that AI generated a submission, the work will receive a grade of zero and will be reported to the Academic Conduct Committee. You may use AI tools to aid your understanding, but all submitted work must be entirely your own.
- Failure to follow these guidelines will result in zero points.

Learning Objective: This homework develops students' ability to synthesize state-space control systems for reference tracking using full-state and output feedback. Students will construct augmented models with integral action, design state-feedback gains via pole placement, and analyze how pole locations influence transient performance and control effort. The assignment further integrates observer design with control synthesis, highlighting the interaction between estimation dynamics and closed-loop behavior. MATLAB and Simulink will be used to implement and evaluate the resulting controllers.

WARNING!!! Matlab assumes that the state-feedback control law is $u = -Kx$ and thus $\tilde{A} = A - BK$. However, we assume $u = Kx$ and thus $\tilde{A} = A + BK$. Therefore, you need to multiply the gains computed by Matlab functions by -1 .

Problem 1. Full state feedback with integrator (FSFI) tracking controller. Consider the system

$$G(s) = \frac{s^2 + s - 6}{(s + 0.1)(s - 1.5)(s^2 + 2)}. \quad (1)$$

1. Augment the system with an integrator state to enable tracking of a constant reference

$$\dot{q} = r - y.$$

Write the augmented system in state-space form. Design a full-state feedback with integrator (FSFI) controller of the form

$$u = K_x x + K_q q,$$

such that the integrator pole is placed at -1 , and the remaining closed-loop poles are chosen to ensure a stable and reasonably fast response (justify your choice). Compute the gains K_x and K_q using `place`.

2. Let $r(t) = 1$ and $x(0) = 0$. Implement the closed-loop system in Simulink using the augmented dynamics. Plot and clearly label all states (including the integrator state), the control input $u(t)$, and the output $y(t)$. Comment on the tracking performance and transient response.
3. Move the integrator pole further into the left half-plane (e.g., -5 , -10) while keeping the remaining poles fixed.
 - How does the settling time change as the integrator pole is moved?
 - How does the control effort (magnitude and aggressiveness of $u(t)$) change?

- What practical factors limit how far the integrator pole can be moved?

Support your answers with simulation results.

Problem 2. Output feedback tracking controller. Consider the system described by (1).

1. Using your results from previous assignments, that is, the observer gain F (from observer design), and the state-feedback gain K_x and integrator gain K_q (from Problem 1), design an observer-based output feedback controller with an integrator. The controller should have the structure

$$\begin{aligned}\dot{\hat{x}} &= A\hat{x} + Bu + F(y - C\hat{x}), \\ \dot{q} &= r - y, \\ u &= K_x\hat{x} + K_qq.\end{aligned}$$

2. Let $r(t) = 1$ and $x(0) = 0$. Simulate the closed-loop system in Simulink. Plot and clearly label the true states $x(t)$, the estimated states $\hat{x}(t)$, the control input $u(t)$, and the output $y(t)$.
3. Compare the performance of the output feedback controller with the full-state feedback controller from Problem 1.
 - How does the transient response differ?
 - How does estimation error affect performance?
 - What role does observer speed play in the overall response?

Support your discussion with simulation results.